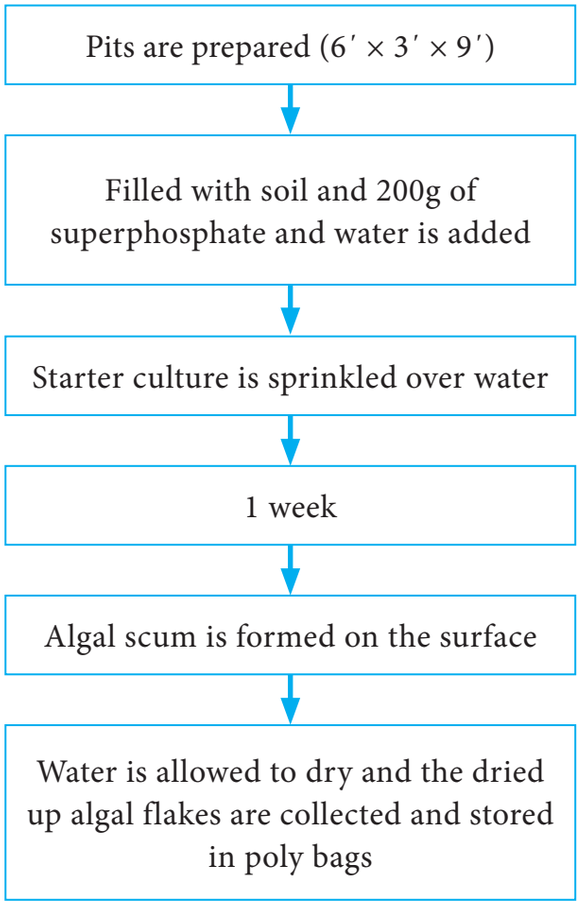
like BG-11 or Pringsheim's media. After several weeks of incubation at 28°C, the individual colonies are picked up, identified and stored. This can be used as starter culture for mass cultivation. Mass culture can be done in 2 ways.

Mass cultivation of BGA

1. Open air shallow culture:



The dried algal flakes around 10kg/ha can be applied in paddy fields after transplantation.

11.2.5 Azolla

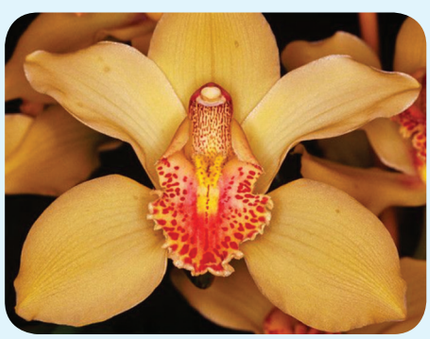
Azolla is a floating freshwater fern. The plant has a branched stem, deeply bilobed leaves which are arranged alternately on the stem and each leaf has a dorsal and ventral lobe (Figure 11.9). The dorsal lobe houses the cyanobacterial symbiont *Anabaena azollae* (Figure 11.10). The fern and the cyanobacteria exhibit symbiotic relationship in which *Anabaena* provides the fern with fixed nitrogen and fern provides niche for the cyanobacteria free from competition from other microorganisms.

Azolla can be used as a nitrogenous biofertilizer for paddy crop. When applied into the paddy fields, *Azolla* provides nitrogen nutrition to standing rice crop and can reduce the need for synthetic fertilizers.

Infobits

Mycorrhiza and orchid germination

In the early stages of their life cycle, all terrestrial orchids are non photosynthetic, totally lacking chlorophyll and relying on carbon(C) acquired from a fungal symbiont (Mycorrhiza) for growth until the production of the first green leaves above the ground, a nutritional strategy termed mycoheterotrophy. Around 200 species of orchids remain achlorophyllous throughout their lifetimes. Species such as *Galeola*, *Gastrodia*, *Corallorhiza*, *Rhizanthella* and many others continue to gain carbon from mycorrhizal fungi.



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